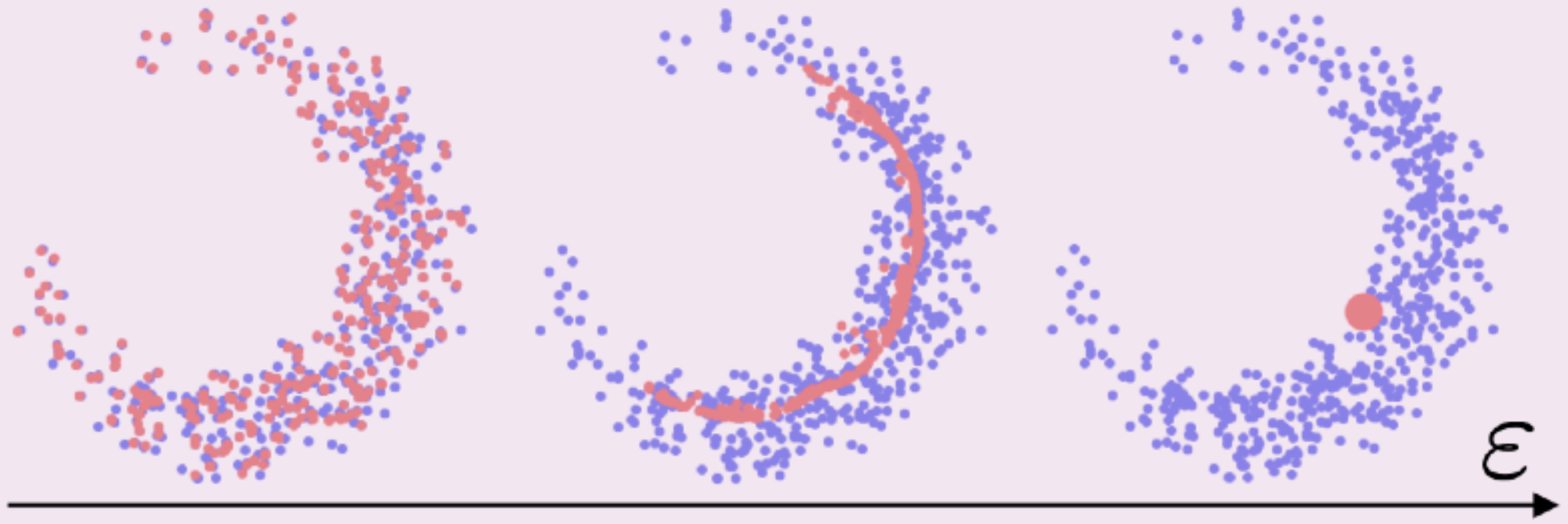
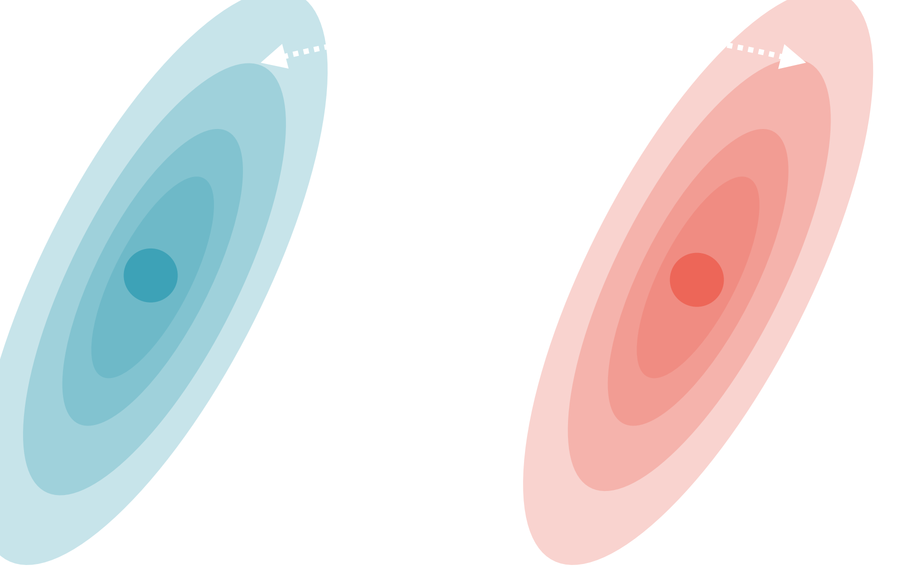
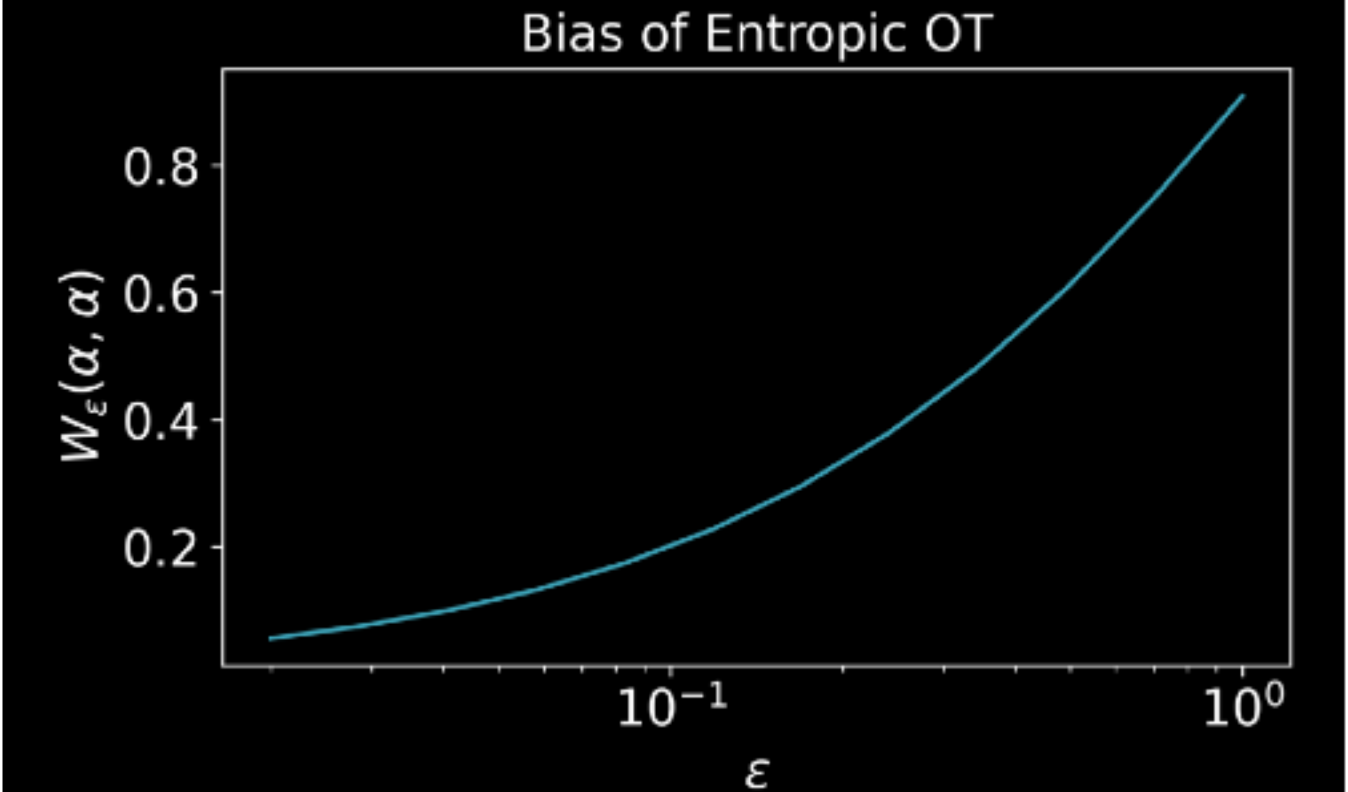
EOT is biased

Problem: $\mathrm{W}_\varepsilon(\alpha, \alpha) \neq 0$

$$\min_{\alpha} \mathrm{W}^p_{\varepsilon,p}(\alpha, \beta)$$

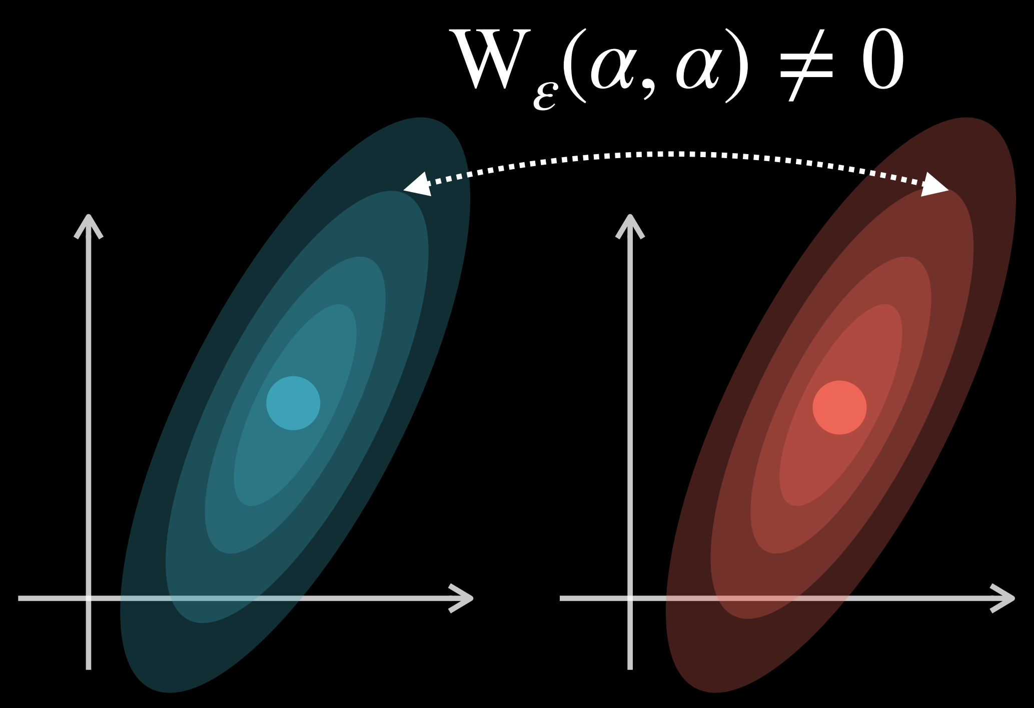


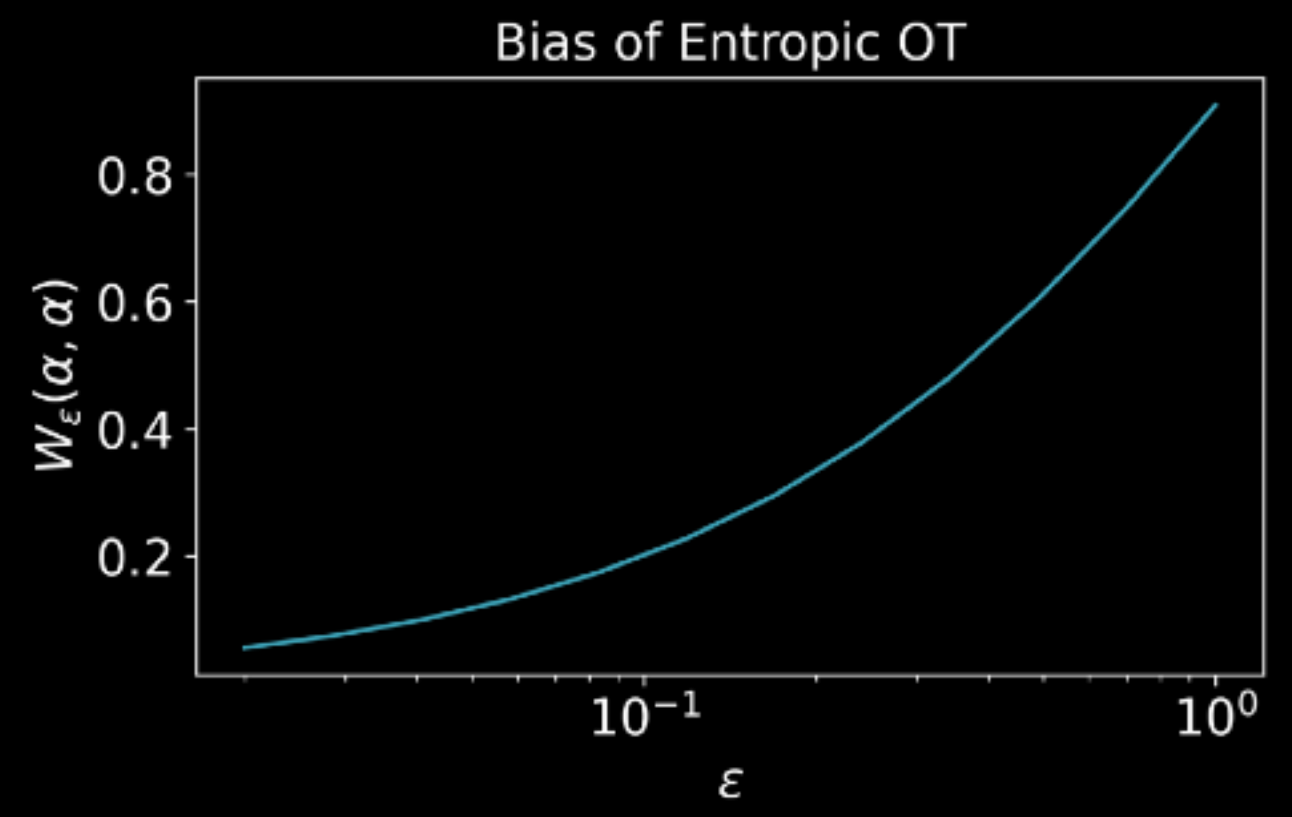



Bias of Entropic OT
0.8
0.6
0.4
0.2
$W_\varepsilon(\alpha, \alpha)$
10^{-1}
10^{0}
ε

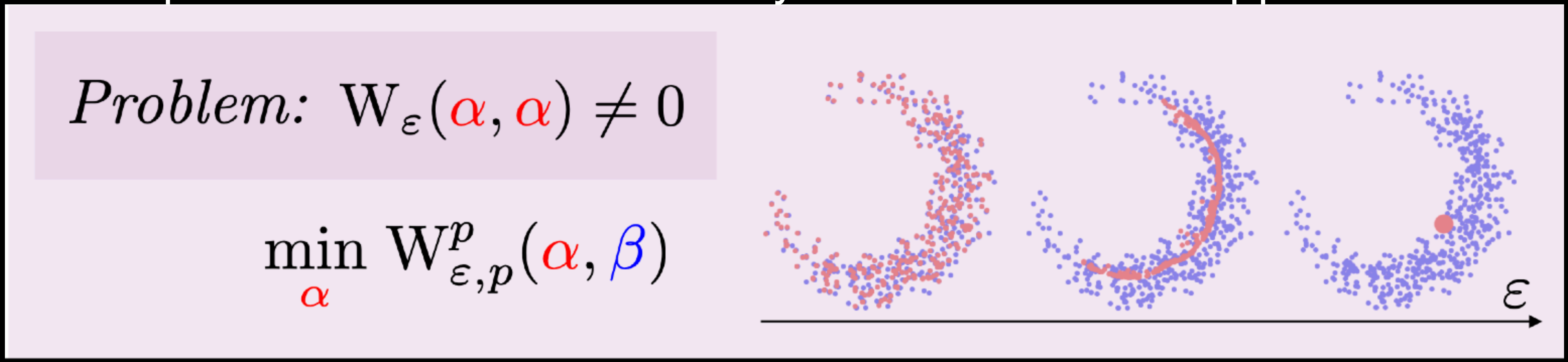
Sinkhorn Divergence: Motivation

- Issue with entropic cost:





- $W_\varepsilon(\alpha, \alpha) > 0$
- This means **EOT is biased** as a dissimilarity measure.
- Consequence: can't be used directly as a "distance" in applications.



Credit: Gabriel Peyre

Debiasing Entropic OT